

Human Development Index (HDI) Predictor

Proposed Solution

Project: Human Development Index (HDI) Predictor

Phase: Project Design Phase

1. Solution Summary

A Flask web application backed by a trained Linear Regression model. Users pick a country and enter its indicator values; the app returns a predicted HDI score and a plain-language development tier, avoiding the need to manually recompute the UNDP composite formula.

2. Key Components

- Dataset generation & EDA — builds a country-indicator dataset and visualizes correlations
- Preprocessing — null-value imputation and label encoding of the Country feature
- Model training — LinearRegression fit on an 90/10 train/test split
- Model serialization — HDI.pkl and country_encoder.pkl saved with Pickle
- Web application — Flask routes for Home, Prediction form, and Result

3. User Benefit

Any user can get an instant, explainable HDI estimate and development classification for a country by entering four numeric indicators — no manual formula computation required.